

Quiz

March 2, 2026

In the "Limited Direct Execution" mechanism, how does the operating system ensure that user programs cannot perform privileged operations (such as I/O requests)?

- ☐ A Allow user programs to execute I/O instructions directly, but the hardware ignores them
- ☐ B Introduce two CPU modes: user mode and kernel mode
- ☐ C Require user programs to request privileged operations via system calls
- ☐ D Use a trap instruction to enter the kernel, perform privileged operations, and then return to user mode
- ☐ E The OS checks each program's instruction set at compile time

提交

Which of the following are TRUE about a Cooperative Approach for the OS to regain control?

- ☐ A The operating system regains control of the CPU when a process makes a system call
- ☐ B A timer interrupt is used to forcibly preempt processes that run for too long
- ☐ C If a process gets stuck in an infinite loop and never makes a system call, the only solution may be to reboot the machine
- ☐ D Processes can explicitly yield the CPU using a system call like yield()
- ☐ E The OS trusts processes to behave reasonably and periodically give up control

提交

多选题 2分

Which of the following state transitions are considered valid?

- ☐ A Running -> Ready
- ☐ B Waiting -> Running
- ☐ C New -> Ready
- ☐ D Ready -> Waiting

提交

Which statements accurately describe the behavior of the fork() system call?

- A** Following a successful fork(), the parent process receives a return value of the PID of the child process, while the child receives 0.
- B** Efficiency is often improved using "copy-on-write" (COW), which delays the actual copying of memory pages until one process attempts to modify them
- C** A process in Linux can be assigned to at most one parent process at the exact same time.
- D** It creates the child process by cloning from the parent process, including the program code and memory, but the program counter is set to the beginning of the program

提交

Which of the followings is/are true about exec()?

- ☐ A When calling exec() to load a new program, the process's open file descriptors are unchanged.
- ☐ B Exec() replaces the user-space memory of the process, but keeps the program counter unchanged.
- ☐ C Exec() changes the PID of the calling process.
- ☐ D When calling exec() to load the same program, the process's memory remains unchanged.

提交

Which of the followings is/are true about context switch?

- ☐ A The operating system can perform a context switch at any arbitrary moment, even when a process is in the middle of executing user code.
- ☐ B Context switch may happen when the kernel takes control of the CPU.
- ☐ C During context switch, the system must save the state of the old process and load the saved state for the new process
- ☐ D Context switch cannot be too frequent, otherwise a high performance overhead may occur.

提交

According to "From L3 to seL4: What Have We Learnt in 20 Years of L4 Microkernels?" which of the following statements about the minimality principle are TRUE?

- A** The minimality principle states that a concept belongs in the microkernel only if moving it outside would prevent implementing required functionality
- B** All L4 kernels to date have achieved a "pure" microkernel design with strict adherence to the minimality principle
- C** The scheduler remains in the kernel in all L4 versions, representing a violation of pure minimality
- D** Source code sizes have grown significantly over 20 years, contradicting the minimality principle
- E** User-level device drivers are a direct consequence of the minimality principle and remain a hallmark of all L4 kernels

提交

主观题 2分

As we learned from "The UNIX Time-Sharing System" paper by Ritchie and Thompson, how is a shell implemented? Describe the sequence of steps the shell takes from reading a command line to executing a command, including the roles of `fork()`, `execute()`, and `wait()`. Also explain how the shell implements I/O redirection (using `<` and `>`) and background processes (using `&`).

提交